



External Sensing in Hybrid Biologically Driven Autonomous Systems

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Background

- Translation of unpredictable biological signals into understandable commands is becoming increasingly useful within the field of medicine and robotics.
- Existing platforms leverage signals ranging from electroencephalography and electromyography to eye-tracking.
- All of these share the same requirements: capture, interpret, and actuate from a stream of noisy biological input.
- **This project aims to implement this pipeline from sense to interpret to actuate by using movement of a goldfish as a variable biological input that turns into real movement. By capturing real-time positions and converting them into vehicle navigation commands, we demonstrate a low cost, accessible framework for studying bio-signal driven control systems acting as proof of concept that can be generalized to different biological inputs.**

Signal Pathway

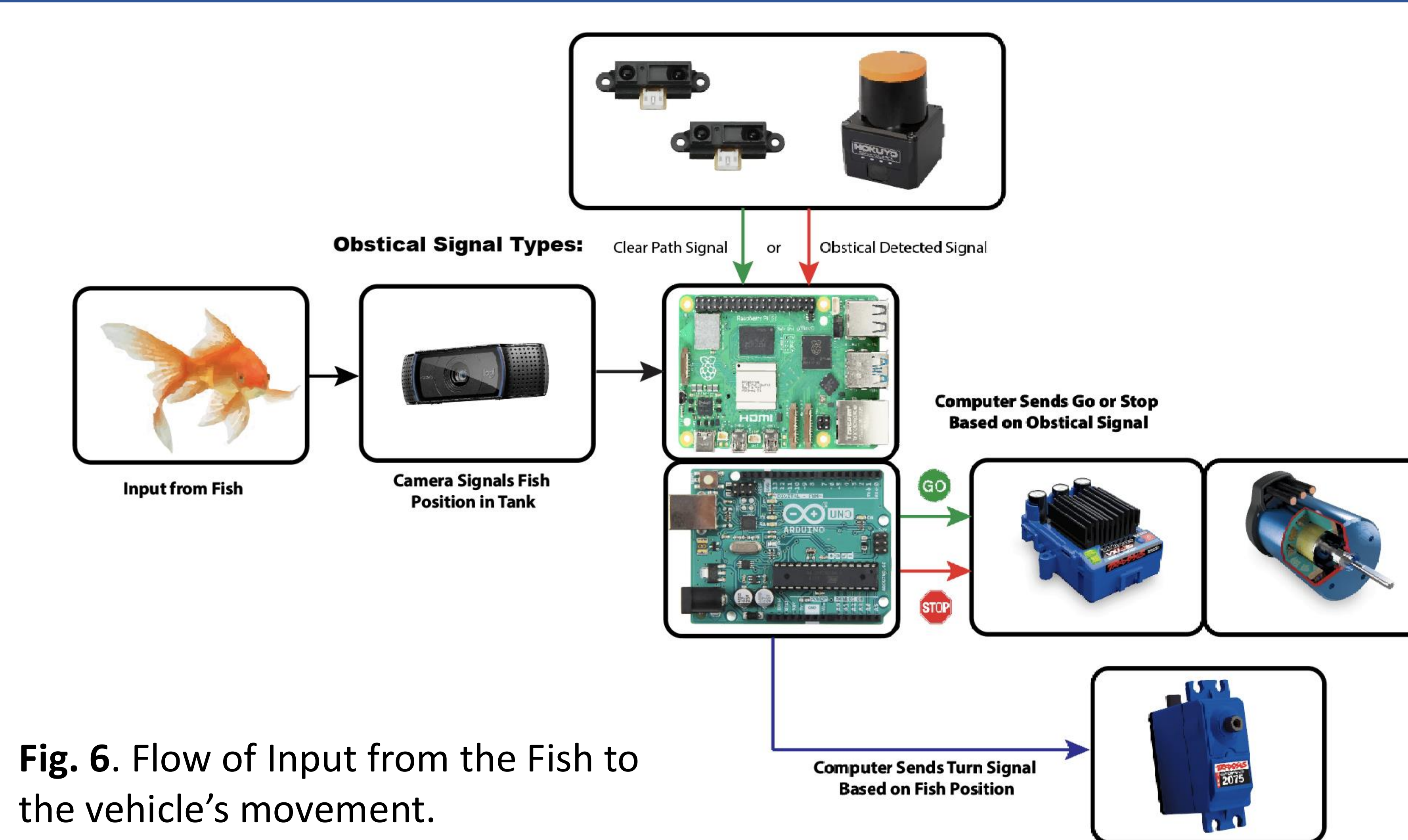


Fig. 6. Flow of Input from the Fish to the vehicle's movement.

Software

Computer Vision (Python):

- Captures live video at 640×480 via OpenCV from a camera
- Converts each frame to HSV color space and applies calibrated thresholding to isolate the fish from the background
- Identifies the largest contour and extracts its centroid as an (x, y) coordinate

Motion Control (Arduino):

- Receives (x, y) coordinates and maps them onto into nine zones
- Outer zones trigger discrete steering angles (60°–120°) and throttle commands scaled to ESC pulse widths (1450–1550 μs)

Safety Override – LiDAR (python), IR (Arduino) :

- LiDAR detects objects in front overriding fish commands and forcing reverse if IR is clear
- IR detects rear objects and stops backward movement

Hardware Design



Fig. 1. Tank Assembly

Tank Assembly

- Custom acrylic tank
- Mounted onto car body
- Waterproofed and sealed
- Houses both the fish and water



Fig. 2. Iteration 1

Iteration 1

- RC car base
- ESC and Servo Integration for speed and direction
- LiDAR mounted to detect objects within 270 degrees
- Camera loosely bound to track fish movement

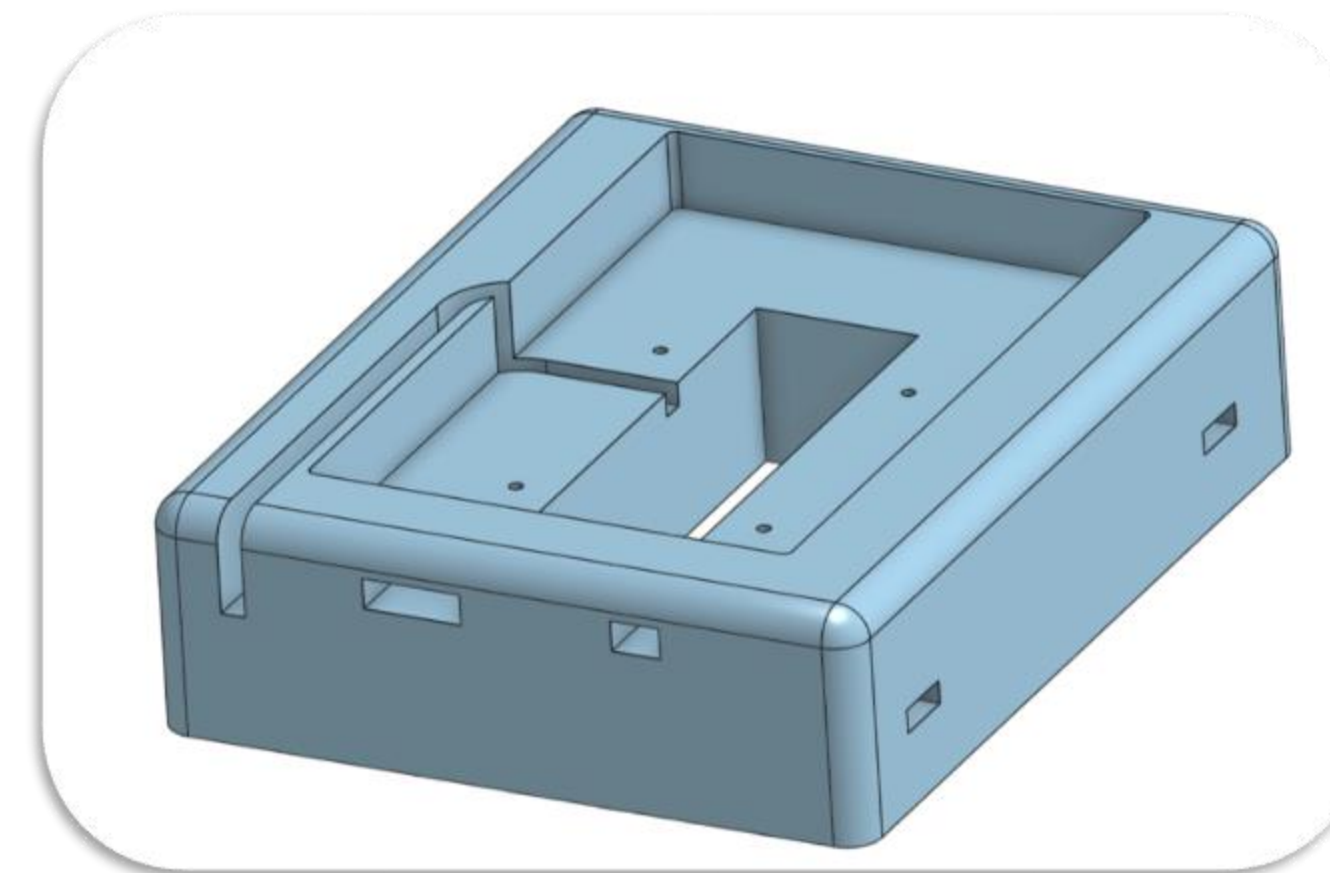


Fig. 3. Touchscreen/Camera Holder

Touchscreen Holder

- Created in Onshape software
- Holds start and stop button on touchscreen
- Secures camera to the top of the tank
- Keeps tank lid in place to avoid leakage

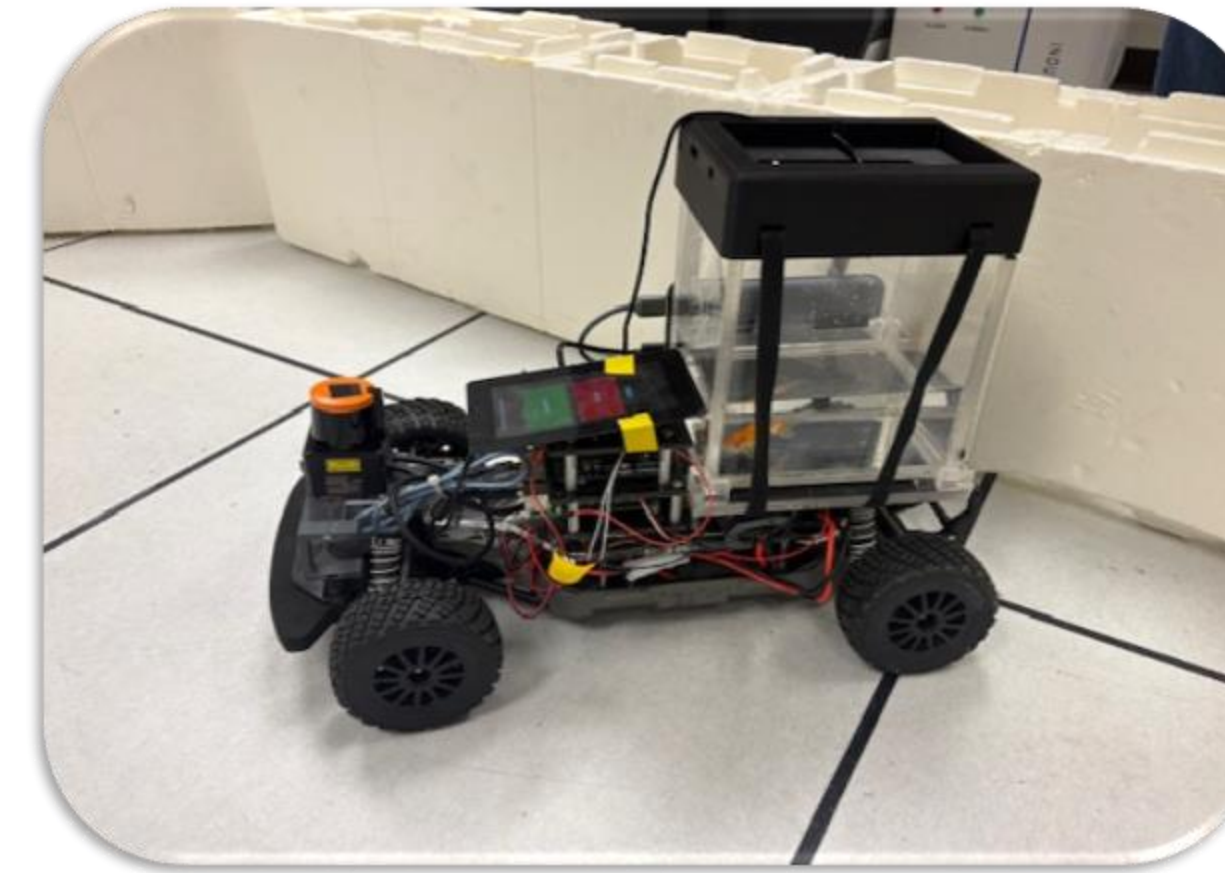


Fig. 4. Final Iteration

Final Iteration

- Camera and touchscreen secure
- Independent of laptop or remote control
- LiDAR and IR sensors for detecting objects
- Drive time of 20 minutes
- Fully self driven

Discussion

- Proved that an imaging-based biological input stream can be captured, interpreted, and actuated on low-cost accessible hardware
- LiDAR safety override confirmed the system handles signal ambiguity without collision
- Swapping the fish for a human visual signal requires changes to the input layer only with the rest of the logic staying identical
- Development of gaze-tracking uses the same concept to allow eye movement to move wheelchairs and robotic arms for people with lack of motor function
- Camera-based eye tracking is also being explored for concussion and neurological assessment, where deviations in real-time gaze patterns are used to detect cognitive impairment

Acknowledgement/References

References:

- [1] Cui et al. (2025). Sage Journals. <https://doi.org/10.1177/10711813251358246>
- [2] Cade et al. (2024). Classification of short and long term mild traumatic brain injury using computerized eye tracking. Scientific Reports, 14, 12686. <https://doi.org/10.1038/s41598-024-63540-8>

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Fig. 5. Demo Video

